Supplementary Material 2

Prompt Development and Templates

In this document, we provide the prompts used in the study and elaborate on the rationale behind their structure and the iterations we went through until arriving at our final prompt. Additionally, we report the results of a descriptive ablation study that examines the sensitivity of our approach to variations in prompt design. The latter is particularly relevant with respect to concerns regarding the validity of LLM-based classification findings in the face of seemingly minor modifications such as changes in component order or the inclusion and exclusion of individual prompt elements.

Rationale and Development

The prompts were developed iteratively, informed by conceptualisations in similar studies (e.g., Bosch et al., 2024; Heseltine & Clemm von Hohenberg, 2024). In an initial version, the prompts contained a short task description and the target comment with context. Naturally, this task description would contain a short description of the project or subreddit context, as well as an explanation of the desired labels and what they refer to. To improve classification quality, we subsequently extended this *zero-shot* approach with context-dependent examples (e.g., galaxy-related examples for the Galaxy Zoo project). We integrated two examples per category, embedded in realistic thread contexts with matching thread starters and previous comments. The resulting prompting technique then employs the principle of *in-context prompting*.

Further, in line with the coding scheme by Wang et al.(2016), we included a decision tree in the prompts, similar to the decision tree that was provided to the human raters. This tree was aimed to guide the model through the classification in a sequential, rule-based manner, evoking chain-of-thought reasoning (Wei et al., 2022). We also experimented with the ordering of the decision tree: an early version checked for *Interactive* before *Constructive (Idea & Elaboration/Reasoning)*, which we later reversed to check *Active (Attention/Engagement & Paraphrase/Repeat)* before *Constructive* and before *Interactive*. This reordering better reflected the hierarchical logic of the ICAP framework, where higher levels of engagement subsume lower ones, and reduced misclassifications where surface-level features (e.g., a reply directed at another user) led the model to assign *Interactive* labels to comments that lacked constructive elements. Finally, to make sure that the model output is consistent, we defined the output format as JSON explicitly at the end of the prompt.

The final structure of the prompt used for the analyses thus consists of general instructions including the role, task and forum contexts, target comment with thread context, the ICAP rules and decision tree, context-specific examples and the output format. This can be seen in Fig. 1.

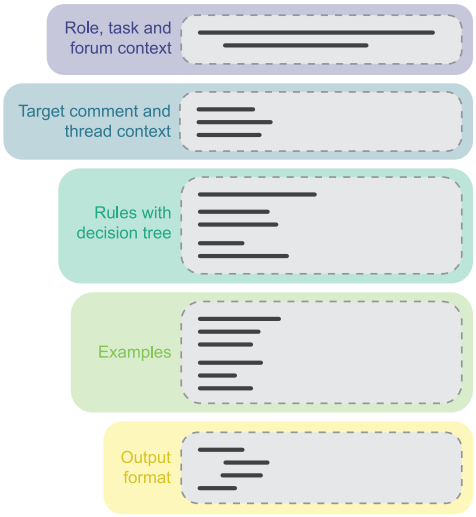


Fig 1. Final prompt structure used for analyses

We note that this development process was guided by qualitative inspection of model outputs against human judgments rather than by systematic optimisation. Changes were motivated by identifiable patterns of misclassification rather than exhaustive search over prompt variants. To provide more rigorous evidence regarding the robustness of the final prompt, we complement this developmental account with a descriptive ablation study.

Ablation Studies

To rule out that classification performance is an artifact of one particular prompt arrangement, we conducted a post-hoc ablation study. Specifically, we removed certain elements and systematically varied the ordering of prompt components to compare the human-model agreement between these different prompt versions.

Arrangement Variants

We start by comparing different variants of arrangement of our final prompt components described above. We fixed the role and task context as the first component and the output format as the last, following established conventions in instruction design where context-setting precedes substantive content and output specifications appear closest to the generation point. This also aligns with findings on positional attention biases in LLMs (Liu et al., 2023). For variant A and B, we thus varied the arrangement of prompt elements in between, as can be seen in Fig. 2.

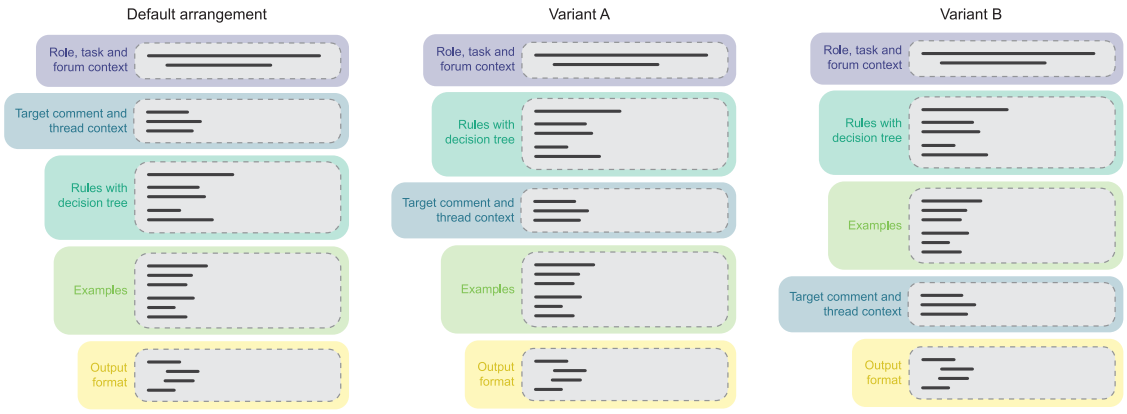


Fig 2. Different arrangements tested. The default arrangements is the one used for the analyses in the paper

We report the results for our *N* = 338 sample of comments, as described in the paper. As can be seen in Table 1, not much variance occurs between different arrangement variants, with both agreement metrics being comparable across variants.

Table 1. Effects of element arrangement

Rater	Measure	Arrangement Variants		
		Default	Variant A	Variant B
Rater 1	*QWK*	.64 [.54,.73]	.64 [.55, .73]	.65 [.56, .74]
	α	.66 [.56,.75]	.65 [.56, .73]	.66 [.58, .74]
Rater 2	*QWK*	.60 [.60,.71]	.58 [.49, .67]	.56 [.46, .66]
	α	.62 [.51,.71]	.60 [.50, .68]	.58 [.48, .67]

Prompt elements

To further test whether the inclusion or exclusion of certain prompt elements leads to different results, we systematically varied how much we include in the prompt, comparing instructing the LLM with rules only, examples only, and none of both. The latter thus means we instruct the LLM only with the role, task and forum context, the target comment and thread context, and the output format. This can be seen in Fig. 3.

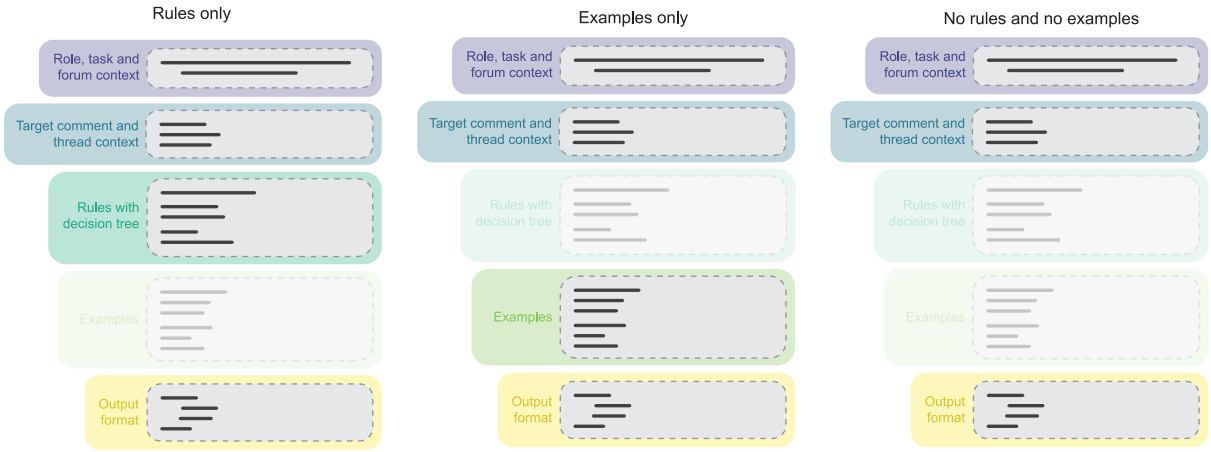


Fig 3. Performance across prompt element arrangements. The default (all elements) is omitted as it corresponds to the results reported above

Table 2 shows the results of these analyses. Expectedly, all tested variants show less agreement with both human raters, with the “no rules and no examples” variant even showing negative agreement values, indicating that the model's ratings systematically diverged from human judgments, performing worse than chance-level agreement.

Table 2. Effects of component ablation

Rater	Measure	Components		
		Rules only	Examples only	No rules; no examples
Rater 1	*QWK*	.55 [.45,.66]	.50 [.40,.60]	-.48 [-.90, -.05]
	α	.56 [.46,.65]	.53 [.42, .63]	-.58 [-.79, -.13]
Rater 2	*QWK*	.48 [.36,.69]	.56 [.46, .66]	-.48 [-.55, -.18]
	α	.49 [.38, .59]	.58 [.48, .67]	-.49 [-.75, .00]

Apart from this, we can see that using only rules leads to higher agreement with rater 1, while using examples only leads to higher agreement with rater 2.

Generally, we can conclude from this ablation study that classification performance is robust to reasonable variations in prompt element arrangement, with negligible differences across ordering variants. At the same time, the component ablation confirms that both rules and examples contribute meaningfully to classification quality, and that their combined inclusion is not incidental but necessary for achieving the agreement levels reported in the main analyses. Together, these findings provide evidence of robustness to the tested prompt variations.

Final prompt template

We now provide the final prompt that was used for the *Galaxy Zoo* project. For the other contexts, only the initial context description and examples changed. Note that the usernames were anonymised.

Here is the final prompt:

You are an expert AI assistant specializing in classifying text comments from online citizen science project discussion forums, where users discuss and classify telescope images of distant galaxies according to their shapes.

Your task is to classify the Target Comment using the ICAP framework categories below.

CONTEXT:

Thread Starter (@userA): "*Looks interesting. Can you be more specific?"*

Previous Comment (@userB): *"Found a potential transit. #transit"*

Target Comment (@userC): "*This is a #goodtransit. The #transiting is visible in all data segments, the dip is U-shaped with a flat bottom, and the star's metrics confirm it's not an eclipsing binary.*"

CLASSIFICATION RULES (apply in order):

1. Off-topic: If a comment lacks content relevance and is irrelevant to the project's scientific content or discussion, label it as Off-topic.
2. Active: If the comment is on-topic, check if it involves paraphrasing, repeating, or using hashtags or technical terms. If so, label it as Active (Paraphrase/Repeat), even when it contains a single hashtag or technical term without further explanation. If the comment simply shows general engagement with project content by seeking information, but without specific paraphrasing or repetition of the discussion's content, label it as Active (Attention/Engagement).
3. Constructive: If the comment is not classified as Active, determine whether it involves constructive behavior. If the user is reasoning, elaborating, explaining or establishing a cause and effect relationship, label it as Constructive (Elaboration/Reasoning). If the user rather introduces a new idea or external content, label it as Constructive (Idea). The latter particularly applies when the comment is the thread starter and contains new ideas, hashtags or technical terms to introduce a topic. Simple questions without the introduction of new content don't qualify here.

4. Interactive: For a comment to be labeled as Interactive, it must be both Constructive and contain references to another user. References include phrases like "I agree with you", "your point makes sense", or other engagement with another user's contributions, which can be explicit (e.g. through a reply or mention) or implicit. Interactive comments reflect a constructive dialogue, where the user builds on, challenges or responds to another user's ideas.

Use this guide to systematically evaluate and label each comment based on the observed cognitive engagement, explaining your answer. Consider the following examples for reference:

CATEGORY EXAMPLES:

0. Off-topic

Example 1: Thread: (@userA): This is one of the most beautiful spiral galaxies I've ever seen! Subject 12345 → Previous: (@userB): Wow, amazing structure. The star-forming regions are so clear. → Target: (@userC): Hi from Arkansas! Label: Off-topic (This comment is a social greeting and is not related to the scientific task of classifying galaxies or the specific discussion about Subject 12345.)

Example 2: Thread: (@userA): I can't seem to load Subject 67890. Is anyone else having trouble? → Previous: (@userB): It's loading fine for me. Maybe try clearing your cache? → Target: (@userC): My laptop can't display the website properly Label: Off-topic (A technical issue related to the user's hardware and the platform, not an engagement with the scientific content (the galaxy image).)

1. Active (Attention/Engagement)

Example 1: Thread: (@userA): New batch of images from the Hubble archive has just been added! → Previous: (@userB): Fantastic! Can't wait to dive in this evening. → Target: (@userC): Starting a new search for distant galaxies. Label: Active (Attention/Engagement) (States an intention to engage with the project's core task, showing general attention and readiness to participate.)

Example 2: Thread: (@userA): The new data release is live! → Previous: Previous comment is the thread starter → Target: (@userB) in reply to @userA: Just saw the new images Label: Active (Attention/Engagement) (Indicates the user has viewed the new project materials (images) without providing further specific detail or analysis.)

2. Active (Paraphrase/Repeat)

Example 1: Thread: (@userA): Look at the faint structures in this one. What a beauty. Subject 55678 → Previous: (@userB): I agree, it looks like there are a lot of arms. → Target: (@userC): This galaxy appears to have so many very faintly visible arms. Its stunning! Label: Active (Paraphrase/Repeat) (Paraphrases the previous user's observation using similar descriptive language, affirming the perception without adding new information.)

Example 2: Thread: (@userA): I think I see a galaxy with a ring structure. Any thoughts? Subject 88990 → Previous: (@userB): Definitely looks like a ring galaxy to me. → Target: (@userC): #rings Label: Active (Paraphrase/Repeat) (Uses a common project hashtag to affirm and categorize the observation made by the previous users, effectively repeating the core idea in a concise format.)

3. Constructive (Idea)

Example 1: Thread: Target comment is the thread starter → Previous: N/A → Target: (@userA): Looks like an interaction @ 7:00 that is extending to the left of the photo ... very interesting ... Subject 23456 Label: Constructive (Idea) (This comment initiates a new thread by pointing out a specific, novel feature (a potential interaction) in a specific location on the image, thereby introducing a new idea for discussion.)

Example 2: Thread: (@userA): What an unusual object. It's so blue and clumpy. Subject 77889 → Previous: (@userB): I've never seen anything quite like it in this project. → Target: (@userC): Could this be a 'Green Pea' galaxy? They are known for being compact and having high rates of star formation. Label: Constructive (Idea) (Introduces a new, specific hypothesis ('Green Pea' galaxy) to explain the observed features, connecting the visual data to an established astronomical category.)

4. Constructive (Elaboration/Reasoning)

Example 1: Thread: (@userA): Are these lines across the image part of the galaxy or just an artifact? Subject 45678 → Previous: (@userB): I've seen things like this before, they are usually artifacts. → Target: (@userC): I think the parallel lines are due to the extreme zooming in, and you are starting to see the light collection patterns of the CCD plate. Label: Constructive (Elaboration/Reasoning) (Provides a detailed explanation for an observation (parallel lines) by proposing a specific cause (CCD patterns from zooming), thereby elaborating on the idea that it's an artifact.)

Example 2: Thread: (@userA): I classified this as a spiral, but I'm not entirely sure. Subject 99001 → Previous: (@userB): It's tricky. What makes you say spiral? → Target: (@userC): The outer arms explain the spiral-like appearance, particularly considering their density and how they curve away from the central bulge. Label: Constructive (Elaboration/Reasoning) (Provides specific reasoning for a classification by linking multiple pieces of evidence (outer arms, density, curvature) to support the 'spiral' conclusion.)

5. Interactive

Example 1: Thread: (@userA): I think this is a satellite trail, not a feature of the galaxy. → Previous: (@userB): @userA I'm not so sure. It seems too wide for a typical satellite. → Target: (@userC) in reply to @userB: I see your point about the width, but it's perfectly straight and has a uniform brightness, which is uncharacteristic of any galactic structure. I think it has to be an artifact. Label: Interactive (Directly addresses @userB, acknowledges their point ('I see your point about the width'), and then presents a counter-argument with specific reasoning ('perfectly straight', 'uniform brightness') to further the dialogue.)

Example 2: Thread: (@userA): Proposing a merger for Subject 33445. The core seems disturbed. → Previous: (@userB): I agree it's disturbed, but it could also be a foreground star overlapping the galaxy's core. → Target: (@userC): That's a good point about the star. Building on your idea, what if we check the SDSS database for an object at that exact coordinate? That would confirm if it's a known foreground star. Label: Interactive (References and validates the previous user's idea ('that's a good point about the star'), then builds upon it by suggesting a concrete next step (checking SDSS), thereby making the interaction collaborative and moving the investigation forward.)

```
OUTPUT FORMAT:
{
  "label": "EXACT_LABEL_NAME",
  "explanation": "Brief reasoning: why this label fits and others don't"
}
```

References

Bosch, N., Williams-Dobosz, D., & Perry, M. (2024). Measuring help-seeking in online course discussion forums with privacy-preserving Large Language Models. In J. Clarke-Midura, I. Kollar, X. Gu, & C. D'Angelo (Eds), *Proceedings of the 17th International Conference on Computer-Supported Collaborative Learning—CSCL 2024*. International Society of the Learning Sciences. https://repository.isls.org//handle/1/10507

Heseltine, M., & Clemm von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text. *Research & Politics*, *11*(1). https://doi.org/10.1177/20531680241236239

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2023). *Lost in the Middle: How Language Models Use Long Contexts* (arXiv:2307.03172). arXiv. https://doi.org/10.48550/arXiv.2307.03172

Wang, X., Wen, M., & Rosé, C. P. (2016). Towards triggering higher-order thinking behaviors in MOOCs. *Proceedings of the Sixth International Conference on Learning Analytics & Knowledge*, 398–407. https://doi.org/10.1145/2883851.2883964

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS '22*, 24824–24837.